COMP11124 Object Oriented Programming

Classes and Objects

Learning Outcomes

In this lab, you will advance your understanding of Python and will learn how to create Python classes and how to use them in your code. You will also learn how to add (and use) methods and attributes to your classes as well as be able to discuss how and why one should modularize and document their code.

Topics Covered

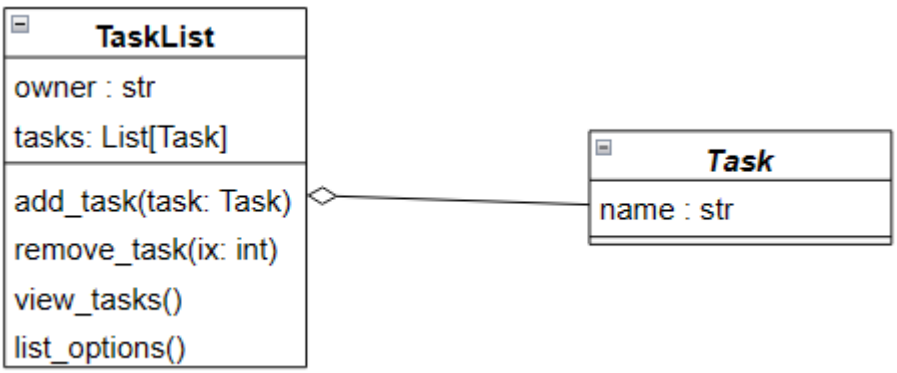
Python: Classes, Methods, Objects, Modularizing code, documentation using docstrings and comments.

Getting Started Task: To retain all the code that you write and be able to work through the exercises in this lab, ***create a new Python file called:*** **`lab_week_4.py`** like how we did it last week. You can include the code of this week here and should continuously execute it to see the output. If you want to see the value of a variable, please *print* it using the **`print()`** function, as I have omitted that in the sample code.

1. Python Classes

Exercise 1: Creating Classes and Initializing Objects.

The last task of last week's class was to create a To Do List program using functions.
This poses ideal to be turned into an OOP program.
Remember, we had a "ToDo list manager" loop that checked for our input, and we could add, remove, and list tasks.
Look at the following UML diagram to see an example of how this functionality could be implemented in an OOP program.



First, we have a class called **`TaskList`**. The top part shows us the Names of the classes, followed by their attributes. In this case, we have a Python list called **`tasks`** that will store all our tasks. We also have a string called **`owner`**, which will store the name of the owner of the task list.

This class handles the task-management functionality. It has four methods: **`add_task, remove_task, view_tasks`** and **`list_options`**:
- The **`add_task`** method takes a parameter called task which should be a **`Task`** object. It then adds the task to the tasks list.

- The **remove_task** method takes a parameter **ix** which is the index of the task that should be removed from the tasks list.
- The **view_tasks** method prints all tasks in the tasks list.
- The **list_options** method prints the options that the user must interact with the task list and handles the user input.

We also have a class called **Task**. To keep it simple, we do not have added methods (yet). We simply have included a task title as an attribute.

If you want to define a class in Python, you use the **class** keyword followed by the name of the class. In our case, this is as follows:

```
class TaskList:
```

We can now add the attributes to the class. To do this, we need to define them in the **__init__** method. This method is always called when we create an object of the class. We can define the attributes in the **__init__** method by using the following code:

```
class TaskList:
    def __init__(self, owner):
        self.owner = owner
        self.tasks = []
```

This then allows us to use the class and create new objects.

```
my_task_list = TaskList("John")
print(my_task_list.owner)
```

Note that we have passed the owner parameter when creating the object. This is because we have defined the owner attribute in the **__init__** method and therefore need to pass a value for it when creating it. If you leave out the owner parameter, you will get an error.
Within the print statement, we have used the dot notation to access the owner attribute of the **my_task_list** object. This is because the owner attribute is an attribute of the **my_task_list** object.
So, if you create a new object of the **TaskList** class, you can access the owner attribute of that object by using the dot notation.

```
someone_else_task_list = TaskList("Jane")
print(someone_else_task_list.owner)
```

Within the **__init__** method we can not only define attributes but also handle other logic such as calling methods. If your task list hypothetically wants to store the owner's name in uppercase, you could do this as follows:

```
class TaskList:
    def __init__(self, owner):
        self.owner = owner.upper()
        self.tasks = []
```

Exercise 2: Adding Methods

Now that we have defined the attributes of the **`TaskList`** class, we can add the methods. Let us start with the **`add_task`** method. As specified, the method takes a parameter called **`task`** which should be a **`Task`** object. It then adds the **`task`** to the **`tasks`** list.

Although we have not yet defined the Task class, we can still build the method. As lists allow us to store any type of data, we can simply assume that the task parameter will be a string (for now).

```
class TaskList:
    def __init__(self, owner):
        self.owner = owner.upper()
        self.tasks = []

    def add_task(self, task):
        self.tasks.append(task)
```

If you want to define a method within a class, you will need to follow the same syntax as when defining a function. The only difference is that you need to add the **`self`** parameter as the first parameter. This is because the method is part of the class and therefore needs to have access to the attributes of the class.
The new method now allows us to add tasks to the **`tasks`** list. We access the object's tasks list by using the dot notation and use the **`self`** parameter to access the current object's tasks list. If you recall, the append method adds an item to the end of a list.

It is very important that you understand the **`self`** parameter. It is always a reference to the current instance of the class and is used to access variables that belong to the class.

We can now add the **`remove_task`** method. This method takes a parameter **`ix`** which is the index of the task that should be removed from the tasks list. We can do this as follows:

```
class TaskList:
    def __init__(self, owner):
        self.owner = owner.upper()
        self.tasks = []

    def add_task(self, task):
        self.tasks.append(task)

    def remove_task(self, ix):
        # Your code here
```

Task: Add the code to the **`remove_task`** method. Hint: You can use the **`del`** keyword to remove an item from a list. For example, if you wanted to remove the item at index 0 from a list called **`my_list`**, you could do this as follows: **`del my_list[0].`**

It is now up to you to add the **`view_tasks`** method. This method should print all tasks in the tasks list.

Task: Add the code to the **`view_tasks`** method. Rather than printing the tasks list directly, you should iterate over the tasks list and print each task individually and also include its index. *Hint*: You can use a for loop to iterate over the tasks list. (*Hint* 2: If you want to be very efficient, read up on the enumerate function in Python https://sparkbyexamples.com/python/for-loop-enumerate-in-python/)

At last, we can add the **`list_options`** method. This method prints the options that the user has to interact with the task list and handles the user input. For this, you can modify your solutions from last week or use the following code as a starting point:

```
    def list_options(self):
        while True:
            print("To-Do List Manager")
            print("1. Add a task")
            print("2. View tasks")
            print("3. Remove a task")
            print("4. Quit")

            choice = input("Enter your choice: ")

            if choice == "1":
                task = input("Enter a task: ")
                self.add_task(task)

            elif choice == "2":
                self.view_tasks()

            elif choice == "3":
                ix = int(input("Enter the index of the task to remove: "))
                self.remove_task(ix)

            elif choice == "4":
                break
```

Exercise 3: Testing the Functionality

Although we do not have fully followed the UML diagram, we have now implemented the functionality of the **`TaskList`** class (albeit we still use a string for the task rather than a Task object).
To check whether you have coded the methods correctly, you can use the following code to create a new **`TaskList`** object and test the functionality in your script:

```
my_task_list = TaskList("YOUR NAME")
# This part is just to test the functionality by adding some tasks to the list
my_task_list.tasks = ["Do Homework", "Do Laundry", "Go Shopping"]
```

```
my_task_list.list_options()
```

Try using the add, view and remove functionality and see whether it works as expected. If you have any problems, you can always look back at the previous examples or ask your lab tutor for help.

Exercise 4: Composition

Composition is a way to combine objects or data types into more complex ones. Here, we are using this to integrate the **`Task`** class into the **`TaskList`** class (as a task is part of a task list).
For now, we have stored the tasks in the tasks list as strings. However, we now want to store them as **`Task`** objects. To be able to create **`Task`** objects, we need to define the **`Task`** class.

Task: Define the **`Task`** class. The class should have one attribute called **`title`**. The **`__init__`** method should take one parameter called **`title`** which should be assigned to the title attribute of that object.

Now that we have defined the class, we can use it in the **`TaskLis`**. To do this, we need to change the choice where we add a task to the tasks list. Instead of adding the task as a string, we need to create a Task object and add it to the tasks list. We can do this as follows in the **`list_options`** method:

```
if choice == "1":
    title = input("Enter a task: ")
    task = Task(title)
    self.add_task(task)
```

Try out the functionality and see whether it works as expected.
You will most likely get an output similar to this:

```
0: <__main__.Task object at 0x000001D8F63B8410>
```

when you try to view the tasks. This is because we have not yet defined a method that returns a string representation of the Task object. We can do this by adding the following method to the Task class:

```
def __str__(self):
    return f"Task: {self.title}"
```

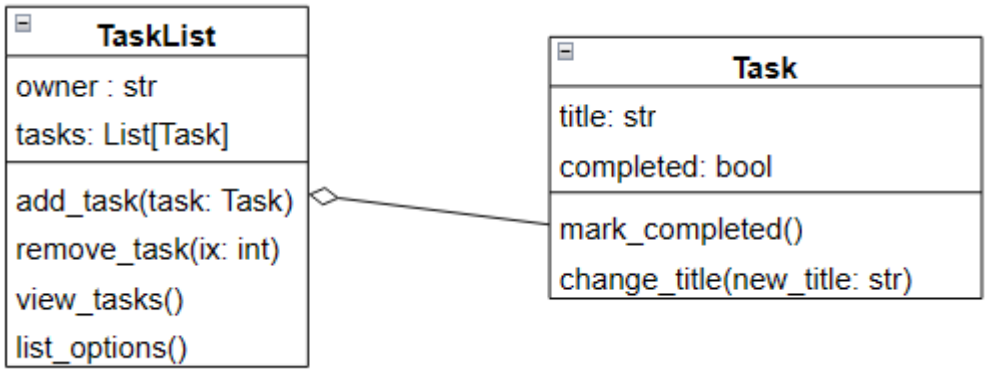
The **`__str__`** method is a special method that returns a string representation of the object. This is useful when we want to print the object or convert it to a string. In the next lecture, we will look at the **`__repr__`** method which is similar to the **`__str__`** method but returns a string representation of the object that is more useful for debugging (but that is a topic for another time).

To test whether everything still works, replace the line of code with **`my_task_list.tasks`** with:

```
my_task_list.tasks = [Task("Do Homework"), Task("Do Laundry"), Task("Go Shopping")]
```

and run it.

Now it is time to add some more functionality.
Look at the following UML diagram to see what functionality we want to add to the Task class:



You will see that we have expanded the **`Task`** class to include an attribute called **`completed`**. This attribute should be a boolean and indicates whether the task has been completed or not. We have also added a method called **`mark_completed`** which should change the completed attribute to True. Additionally, we have also added a method **`called change_title`** which should take a parameter called **`new_title`** and change the **`title`** attribute to the new title.

Task: Change the code in the Task class accordingly and test the functionality.
You will need to add the completed attribute to the **`__init__`** method and set it to False. You will also need to add the **`mark_completed`** method and the **`change_title`** method to the **`Task`** class, with functionality as described above (and do not forget to add the **`completed`** attribute to the task's **`__str__`** method).

Task: Now that we have added the functionality to the **`Task`** class, we can provide options for the user to utilize this functionality. For this, you will need to modify the **`list_options`** method in the **`TaskList`** class.
Add the following options to the **`list_options`** method:
- Mark a task as completed
- Change the title of a task

Note: You will need to add additional if statements to the **`list_options`** method to handle the new options.

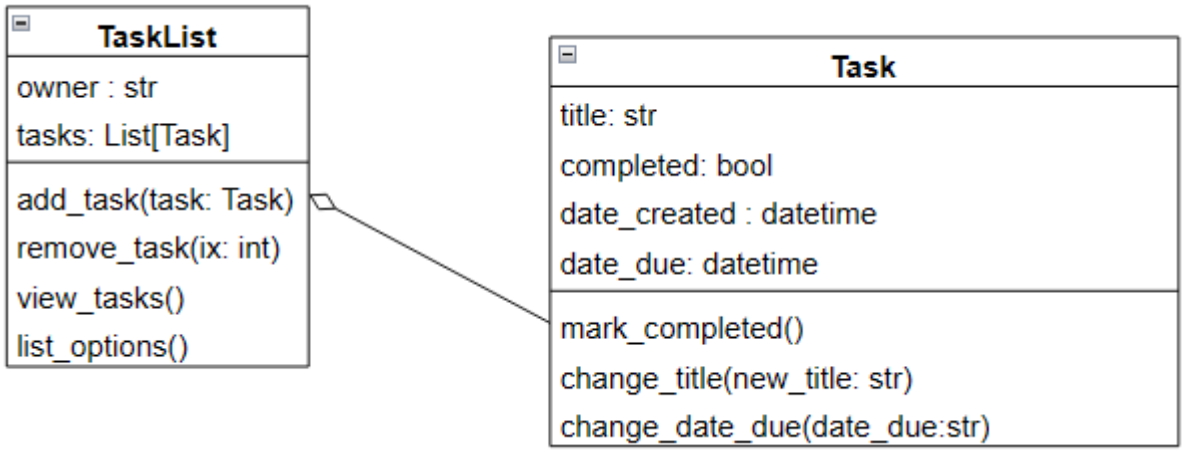
2. Python Libraries

Python libraries are collections of functions and methods that allow you to perform actions, without having written the code yourself. Several are included in the standard Python installation, such as the math or datetime libraries.

Exercise 1: Adding Dates

What would a to-do list be without dates? Let's add due dates and created dates to our tasks.

To do this, we change the Task class as in the following class diagram:



You will see that we have added two attributes to the **`Task`** class: **`created`** and **due**. The created attribute should store the date and time when the task was created. The **due** attribute should store the date and time when the task is due. Both attributes should be datetime objects. Additionally, we also add a **`change_date_due`** method which should take a parameter called **`date_due`** and change the due attribute to the new date due.

One option would be to store any date as a string. However, this would make it difficult to perform any calculations with the dates. For example, if we wanted to calculate how many days are left until the due date, we would need to convert the string to a **`datetime`** object first. Luckily, Python has a library called **`datetime`** that allows us to work with dates and times.

To use the **`datetime`** library, we need to import it first. Now, it may be tempting to put the following code statement only where we need it, but it is good practice to put all **`import`** statements at the top of your script. This makes it easier to see which libraries you are using in your script.
Therefore, add the following import statement to the top of your script:

```
import datetime
```

Now that we have imported the **`datetime`** library, we can use it to create **`datetime`** objects. We can do this by using the **`datetime`** class from the **`datetime`** library. We can then use the **`datetime`** class to create a **`datetime`** object. For example, if we wanted to create a **`datetime`** object for the 1st of January 2021 at 12:00, we could do this as follows (this is an example, but does NOT need to be added to your script):

```
date = datetime.datetime(2021, 1, 1, 12, 0)
date = datetime.datetime.strptime("2021-01-01 12:00", "%Y-%m-%d %H:%M")
```

The latter example uses the **`strptime`** method from the datetime library. This method takes two parameters: the date as a string and the format of the date. You can find examples of this format here: https://www.w3schools.com/python/python_datetime.asp

Change the code in the Task class accordingly and test the functionality.
First, change the **`__init__`** method to include the **`date_due`**:

```
def __init__(self, title, date_due):
```

and then add the attributes to the class:

```
self.date_created = datetime.datetime.now()
self.date_due = date_due
```

Now, you should also change the **`TaskList`** class to use the functionality.

First, change the option where we add a task to the tasks list. Instead of adding the task as a string, we need to create a **`Task`** object with **`title`** and **`date_due`** attributes and add it to the tasks list. We can do this as follows in the **`list_options`** method (note, replace the code that you have added previously):

```
            if choice == "1":
                title = input("Enter a task: ")
                input_date = input("Enter a due date (YYYY-MM-DD): ")
                date_object = datetime.datetime.strptime(input_date, "%Y-%m-%d")
                task = Task(title, date_object)
                self.add_task(task)
```

Task: In the same format as above, add the option to change the due date of a task. It is most likely that in your user options, you will need to add an additional option for editing a task, where you can group the option to change the title and the due date.

Task: Once you have added the functionality to change the due date of a task, please test the functionality. You want to ensure that your code works as expected. For example, if you change the due date of a task, you want to ensure that the due date is actually changed and that you can add or edit the tasks.

3. Modularizing your Code

Exercise 1: Restructuring

As discussed during the lecture, modularizing your code is an important aspect of programming. It allows you to break down your code into smaller, more manageable chunks. This makes it easier to read and understand your code and also makes it easier to extend your code in the future.

Modularizing our code allows us to have a single script that contains the main functionality of our program. This is called the **`main`** script. This is our entry point to the program and is the script that we run to start the program.
We can then move the **`Task`** and **`TaskList`** classes into separate scripts (e.g. files), called modules. These modules can then be imported into the main script, tying the functionality together.
Whilst just now, it may seem like a lot of work to modularize your code, it will pay off in the long run.

Task: Create a new folder called ToDoApp wherever you want to store your code. In this folder, create a new file called main.py. This will be our main script. Then create two new files called **`tasks.py`** and **`task_list.py`**. These will be our modules.
Fill the **`tasks.py`** file with the **`Task`** class and the **`task_list.py`** file with the **`TaskList`** class code

Do not forget to import the **datetime** library in both modules. When you fill the **TaskList** you may also see a squiggly line under the **Task** class. This is because the **Task** class is not yet defined in the **TaskList** module. We will fix this by importing the **Task** class into the **TaskList** module.
Add the following import statement to the top of the **task_list.py** file:

```
from tasks import Task
```

At the end, your file structure should look like this:
ToDoApp/
  main.py
  tasks.py
  task_list.py

Exercise 2: Main()

Now let us populate the main entry point of our program, the main.py file.
First, we need to import the **TaskList** class from the modules. (Note: we do not need to import the Task class as it is not directly used here) Then we can build a **main()** function that contains the main functionality of our program.

This is followed by only executing the **main()** function if the script is run directly. This is important because if we import the main.py file into another script, we do not want the **main()** function to be executed. Therefore, we place the call to this function into a conditional:

```
from tasklist import TaskList

def main():
    task_list = TaskList("YOUR NAME")
    task_list.list_options()


if __name__ == "__main__":
    main()
```

Task: Test whether your code works as expected. You should be able to run the **main.py** file and see the options to interact with the task list.

In OOP we want to ensure that our classes are as independent as possible. This means that we want to avoid having to change multiple classes if we change something in one class. A lot of the ToDo logic is however still within the **TaskList** class within the **list_options** method.
Let us take those instructions out of the **TaskList** class and move them into the **main.py** file. The rationale for this is that whilst the **TaskList** class is responsible for managing the tasks, it is not responsible for the user interaction. This could be the responsibility of another class (maybe one that builds a GUI), but for now, we will simply move it into the **main.py** file.

To do this, copy all the code from the **list_options** method in the **TaskList** class into the **main()** function in the main.py file. Then, remove the **list_options** method from the **TaskList** class. Ensure that in the **main()** function you have fixed the indentation of the code that you have copied over.
Then remove the line:

```
    task_list.list_options()
```

as we do not need to call this method anymore (since we moved the logic).

If you are using VS Code you may see a few yellow squiggly lines under the code that you have copied over. This is because the code is now outside of the **TaskList** class and therefore does not have access to the **self** parameter.
For this, we can employ a very easy fix. Simply replace all instances of self. with **task_list**. (or whatever you have called your **TaskList** object). This will ensure that the code still works as expected.
(Note: If you are using VS Code you can simply double click on the **self** keyword and then press CTRL + F2 to select all instances of the **self** keyword. You can then simply type **task_list**. and it will replace all instances of self with **task_list**.)

Now you will need to import the required datetime and Task classes:

```
from tasks import Task
import datetime
```

To allow us to test the program without always manually adding tasks, we can add some code to the **main()** function that adds some tasks to the task list.

Copy the following code into the **main.py** file:

```
def propagate_task_list(task_list: TaskList) -> TaskList:
    """Propagates a task list with some sample tasks.

    Args:
        task_list (TaskList): Task list to propagate.

    Returns:
        TaskList: The propagated task list.
    """
    task_list.add_task(Task("Buy groceries", datetime.datetime.now() - datetime.timedelta(days=4)))
    task_list.add_task(Task("Do laundry", datetime.datetime.now() - datetime.timedelta(days=-2)))
    task_list.add_task(Task("Clean room", datetime.datetime.now() + datetime.timedelta(days=-1)))
    task_list.add_task(Task("Do homework", datetime.datetime.now() + datetime.timedelta(days=3)))
    task_list.add_task(Task("Walk dog", datetime.datetime.now() + datetime.timedelta(days=5)))
    task_list.add_task(Task("Do dishes", datetime.datetime.now() + datetime.timedelta(days=6)))

    return task_list
```

Additionally, add the following code to the **main()** function in the line below the while True statement:

```
    # propagate the task list with some sample tasks
```

```
    task_list = propagate_task_list(task_list)
```

This allows us to test the program without having to manually add tasks to the task list.

Task: Whilst the actual functionality of the program has not changed, we have now modularized our code further. To ensure that everything still works as expected, test your code again. You should be able to run the **`main.py`** file and see the options to interact with the task list (but this time the logic mainly happens in the **`main.py`** file rather than in the **`TaskList`** class).

4. Type Checking and Documenting your Code.

Exercise 1: Type Checking

Type checking is an important aspect of programming. It allows you to ensure that the data you are using is of the correct type. For example, if you want to add two numbers together, you need to ensure that both numbers are of the type int or float. If you try to add a string to an int, you will get an error. Other programming languages such as Java or C++ require you to specify the type of a variable when you define it. This allows us to catch errors early on. Python on the other hand is a dynamically typed language. Dynamically typed languages do not require you to specify the type of a variable when you define it. This allows you to be more flexible but also means that you need to be more careful when using variables.
However, it is very good practice to add type hints to your code. Type hints allow you to specify the type of a variable or function parameter. This makes it easier to use an IDE such as VS Code to check whether you are using the correct type (since it will show you an error if you are not). It also makes it easier for other people to understand your code.

You will have seen examples of type checking in the lecture already. Here is a very good small cheat sheet that you can refer to if you are unsure how to add type hints to your code:
https://mypy.readthedocs.io/en/stable/cheat_sheet_py3.html

Generally, the syntax is as follows:

```
variable_name: type
```

This can be added to the variable definition or any function/method parameter. For example, if we wanted to add type hints to the Task class, we could do this as follows:

```
class Task:
    def __init__(self, title: str, date_due: datetime.datetime):
    # remaining code
```

What we specify here is that the title parameter should be a string and the **`date_due`** parameter should be a **`datetime.datetime`** object.

We can also specify custom types. For example, in our **`tasklist`** in the **`add_task`** method, we want to ensure that the task parameter is a Task object. We can do this as follows:

```
    def add_task(self, task: Task) -> None:
    # more code.
```

You may also see the arrow syntax **->** . This is used to specify the return type of a function or method. In this case, we specify that the **add_task** method does not return anything (**None**).

In some cases, we have a method that will return something, such as the **__str__** method in the Task class. We can specify this as follows:

```
def __str__(self) -> str:
    # more code.
```

Task: Ensure that all your classes and functions have type hints. If you are unsure about the type of a variable, you can always use the **type()** function to check the type of a variable.
Therefore, add type hints for the return types of all methods and functions and for all parameters and variables where possible:
- all methods in the **Task** class
- all methods in the **TaskList** classs
- the main function in the **main.py** file

Please Note: Depending on your computer setup, the follow part may not work smoothly. As type-checking before running code is optional (but recommended) you can skip this if you experience an error.

To allow us to check whether we have added type hints correctly, we can use a tool called **mypy**. This is a static type checker for Python. Follow the guide here: https://mypy.readthedocs.io/en/stable/getting_started.html to install **mypy** on your computer. If you are using your own PC, you only need to do this once. If you are using the UWS computers, you will need to do this every time you work on a new computer/class.

To use **mypy**, you simply need to run the following command in the terminal (at the bottom of VS Code):

```
mypy --strict main.py
```

This will check whether you have added type hints correctly. If you have not, you will see error messages telling you where you have made a mistake. You can then go back to your code and fix the errors.
If everything is correct, you will see something akin to this:

```
acticals\ToDoApp>mypy --strict main.py
Success: no issues found in 1 source file
```

Exercise 2: Docstrings and Comments

Another important aspect of programming is to document your code. This makes it easier for other people to understand your code and makes it easier for you to understand your code if you have not looked at it for a while.
There are two main ways of documenting your code: docstrings and comments.

For docstrings, it may be worth reading the first part of this guide: https://peps.python.org/pep-0257/ to understand what they are and how they are used.

Docstrings are used to document classes, functions, and methods. They are placed right after the definition of the class, function or method and are enclosed in triple quotes. For example, if we wanted to add a docstring to the Task class, we could do this as follows:

```
class Task:
    """Represents a task in a to-do list. <-- this is a class docstring.
    """

    def __init__(self, title: str, date_due: datetime.datetime):
        """Creates a new task. <-- this is a method docstring.

        Args:
            title (str): Title of the task.
            date_due (datetime.datetime): Due date of the task.
        """

    # some other code

    def mark_completed(self) -> None:
        """Marks the task as completed."""

        self.completed = True
```

Generally, the information that you should include in a docstring is:
- a description of what the class, function or method does
- a description of the parameters and their types
- a description of the return type (if applicable)

There is no set format for docstrings, but it is important that you are consistent and that you include all the information that is required to understand the class, function or method. If in doubt, use the example above as a starting point.

Task: Add docstrings to all classes, functions and methods in your code. Ensure that you have included all the information that is required to understand the class, function or method. You do not need to add docstrings to the main.py file or the **`__init__`** and **`__str__`** methods of the **`Task`** and **`TaskList`** classes.

Another form of documentation are comments. Comments are used to explain parts of your code. They are generally used to explain why you have done something in a certain way. For example, if you have used a certain algorithm, you may want to explain why you have used this algorithm.
Comments are created by using the # symbol. Anything after the # symbol is ignored by Python. For example, if we wanted to add a comment to the main function in the main.py file, we could do this as follows:

```
        if choice == "1":
            title = input("Enter a task: ")
            input_date = input("Enter a due date (YYYY-MM-DD): ")
            date_object = datetime.datetime.strptime(input_date, "%Y-%m-%d")
            # create a new task object based on the title entered and the date entered
            task = Task(title, date_object)
            task_list.add_task(task)
```

You should use comments sparingly and only use them to explain why you have done something in a certain way or it is not obvious what a certain part of your code does.

5. Portfolio Exercises

The following portfolio exercises will form part of the coursework portfolio you will need to submit. There will be no solutions provided to those exercises after the class, but you should complete them to the best of your ability.

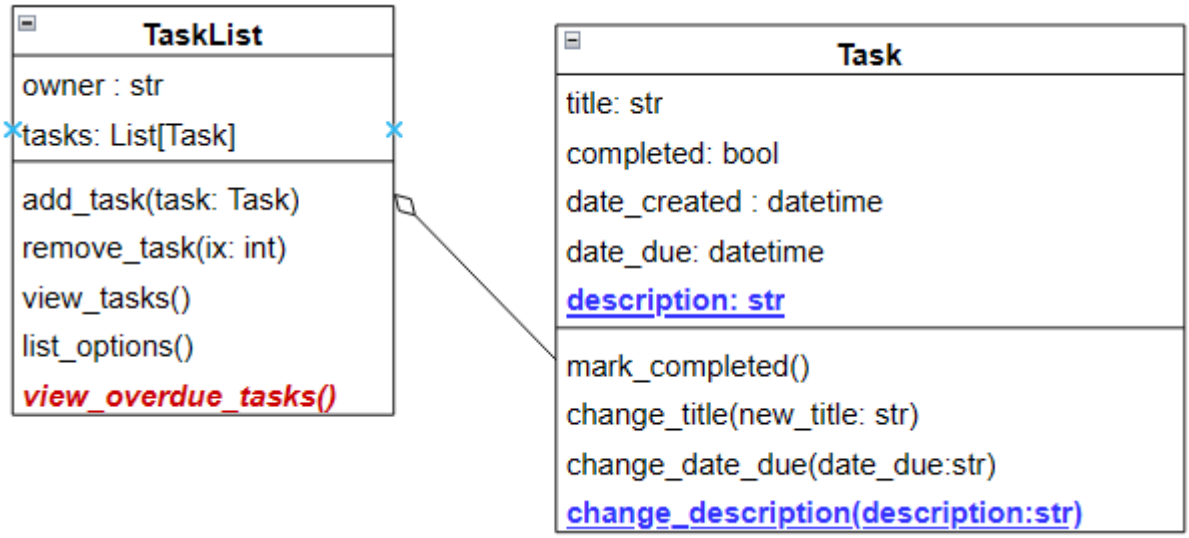
Portfolio Exercise 1

Add a **`description`** attribute to the Task class. This should be a string that describes the task but is entirely optional for the user of your program to provide. For this, you should:
- add the **`description`** attribute to the **`Task`** class and allow for it to be passed as a parameter to the **`__init__`** method.
- add a method called **`change_description`** that allows the user to change the description of a task
- change the **`__str__`** method to include the description of the task
- change the **`main()`** function to allow the user to change the description of a task in choice 4 where the user can also change the title and due date of a task

Portfolio Exercise 2

Add a method to the **`TaskList`** class that allows the user to view all overdue tasks. For this, you should:
- add a method called **`view_overdue_tasks`** that prints all tasks that are overdue based on the current date.
- change the **`main()`** function to allow the user to view all overdue tasks in an additional choice

A visualisation of this is included in the following UML diagram, with Portfolio Exercise 1's additions being <u>underlined</u> and Portfolio Exercise 2's additions being *italic.*


TaskList
owner : str
tasks: List[Task]
add_task(task: Task)
remove_task(ix: int)
view_tasks()
list_options()
view_overdue_tasks()
Task
title: str
completed: bool
date_created : datetime
date_due: datetime
description: str
mark_completed()
change_title(new_title: str)
change_date_due(date_due:str)
change_description(description:str)